

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: CSA1603

COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

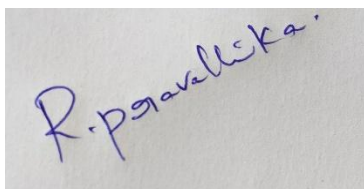
Criteria	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total(10)
Weightage	20%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I RAJANEDI DATI PRAVALLIKA/(192525473) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

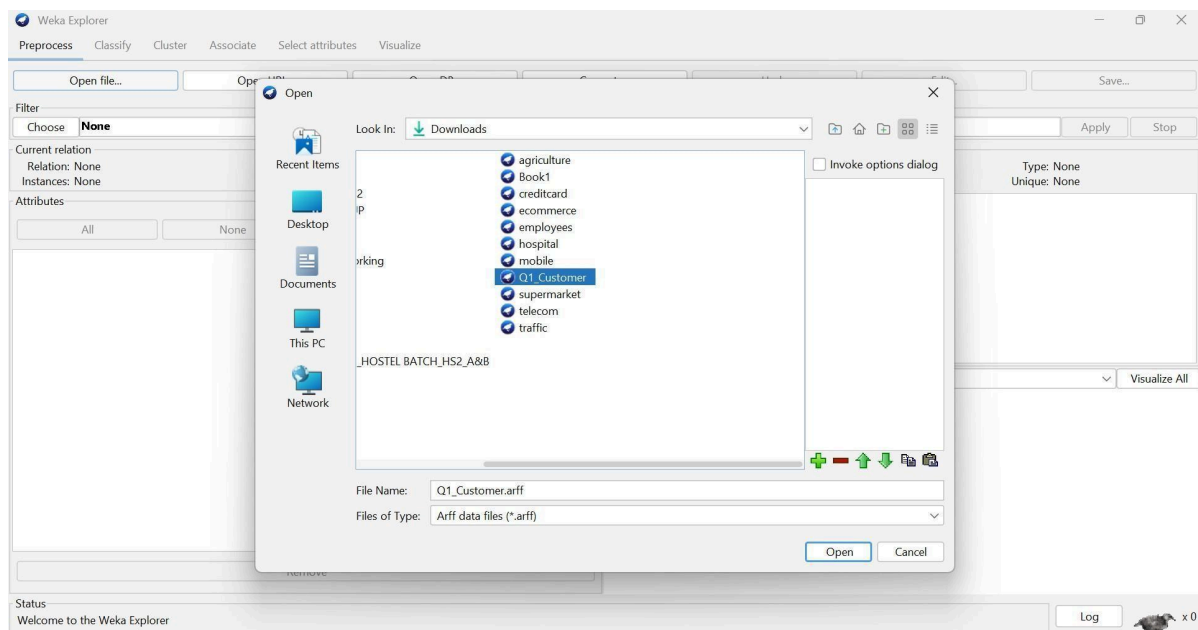
Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalCluster** - N 2 - L SINGLE - P - A "weka-core EuclideanDistance - R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalCluster

Cluster output

Initial starting point(s) view(s):

Cluster 0: C0,28000,45

Cluster 1: C1,60000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (2.0)
Customer	C1	C1	C2
Income	48000	34000	42500
SpendingScore	54	45	37.5

Time taken to build model (full training data) : 0.02 seconds

== Model and evaluation on training set ==

Clustered Instances

0	2 (60%)
1	2 (40%)

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalCluster** - N 2 - L

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split

☐ Classes to clusters evaluation (Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalCluster

Weka Cluster Visualizer: 06:38:09 - SimpleKMeans (customer_clustering)

X: Instance_number (Num)

Y: Customer (Nom)

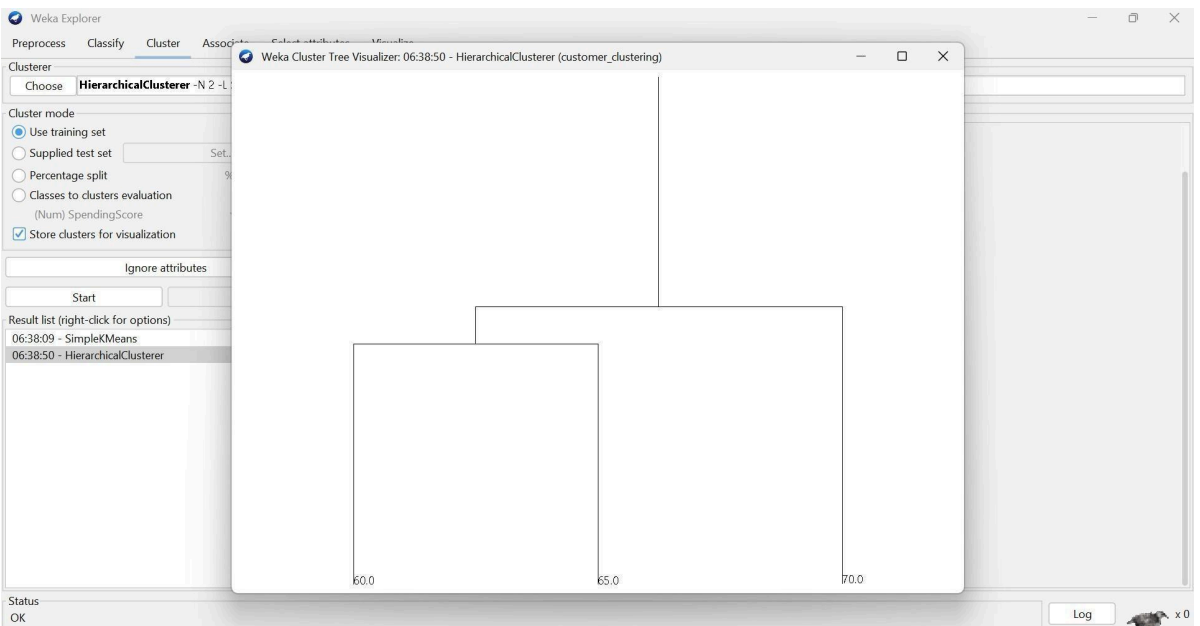
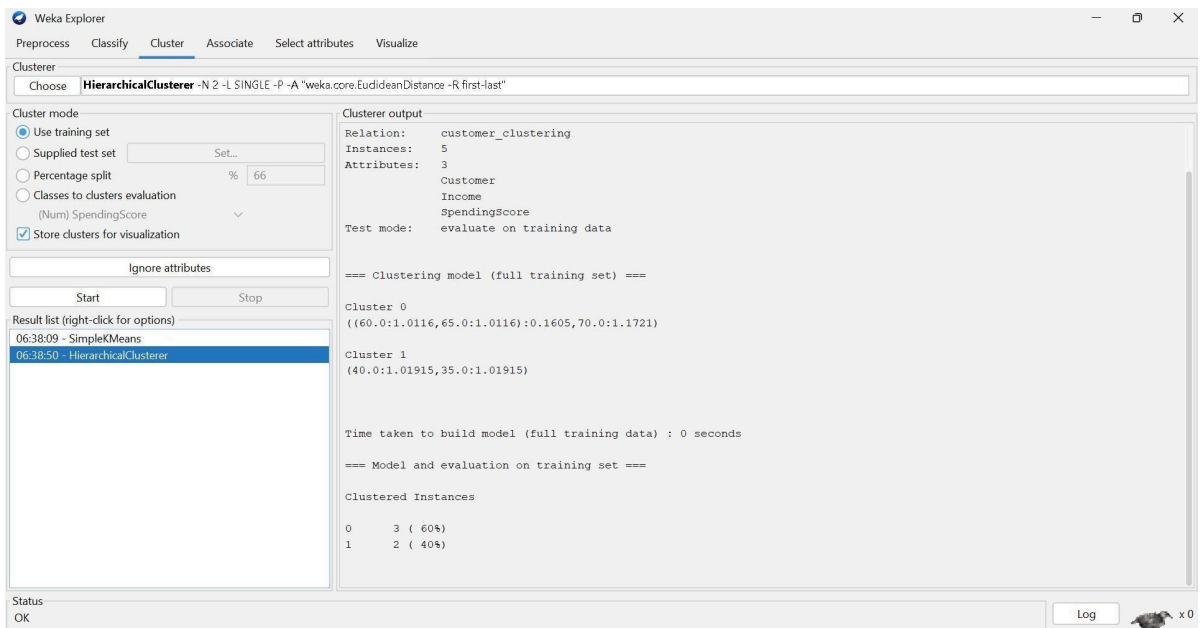
Colour: Cluster (Nom)

Select Instance

Plot: customer_clustering_clustered

Class colour

Status OK Log x0



Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

OUTPUT:

The top screenshot shows the Weka Explorer interface with the 'Open' dialog box open. The 'Look In' dropdown is set to 'Downloads'. The file list shows various files, with 'Q2_Document.arff' selected. The 'File Name' field contains 'Q2_Document.arff' and the 'Files of Type' dropdown is set to 'Arff data files (*.arff)'. The 'Open' button is highlighted.

The bottom screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'SimpleKMeans'. The 'Cluster mode' section has 'Use training set' selected. The 'Ignore attributes' section is empty. The 'Result list' shows '06:43:03 - SimpleKMeans' selected. The 'Clusterer output' section displays the following information:

```
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute  Full Data  Cluster#
              (5.0)    (2.0)    (3.0)
-----
Document   D1      D3      D1
Word1      0.41   0.775   0.1667
Word2      0.35   0.125   0.5
Word3      0.24   0.1     0.3333

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances
0      2 ( 40%)
1      3 ( 60%)
```

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Clusterer

Choose

EM -I 100 -N -1 -X 10 -max -1 -ll -cv 1.0E-6 -ll -iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

Set...

%

66

(Num) Word3

☒ Store clusters for visualization

Ignore attributes

Start

Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

06:43:27 - EM

Cluster output

D4 2

D5 2

[total] 10

Word1

mean 0.41

std. dev. 0.3007

Word2

mean 0.35

std. dev. 0.1949

Word3

mean 0.24

std. dev. 0.12

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 5 (100%)

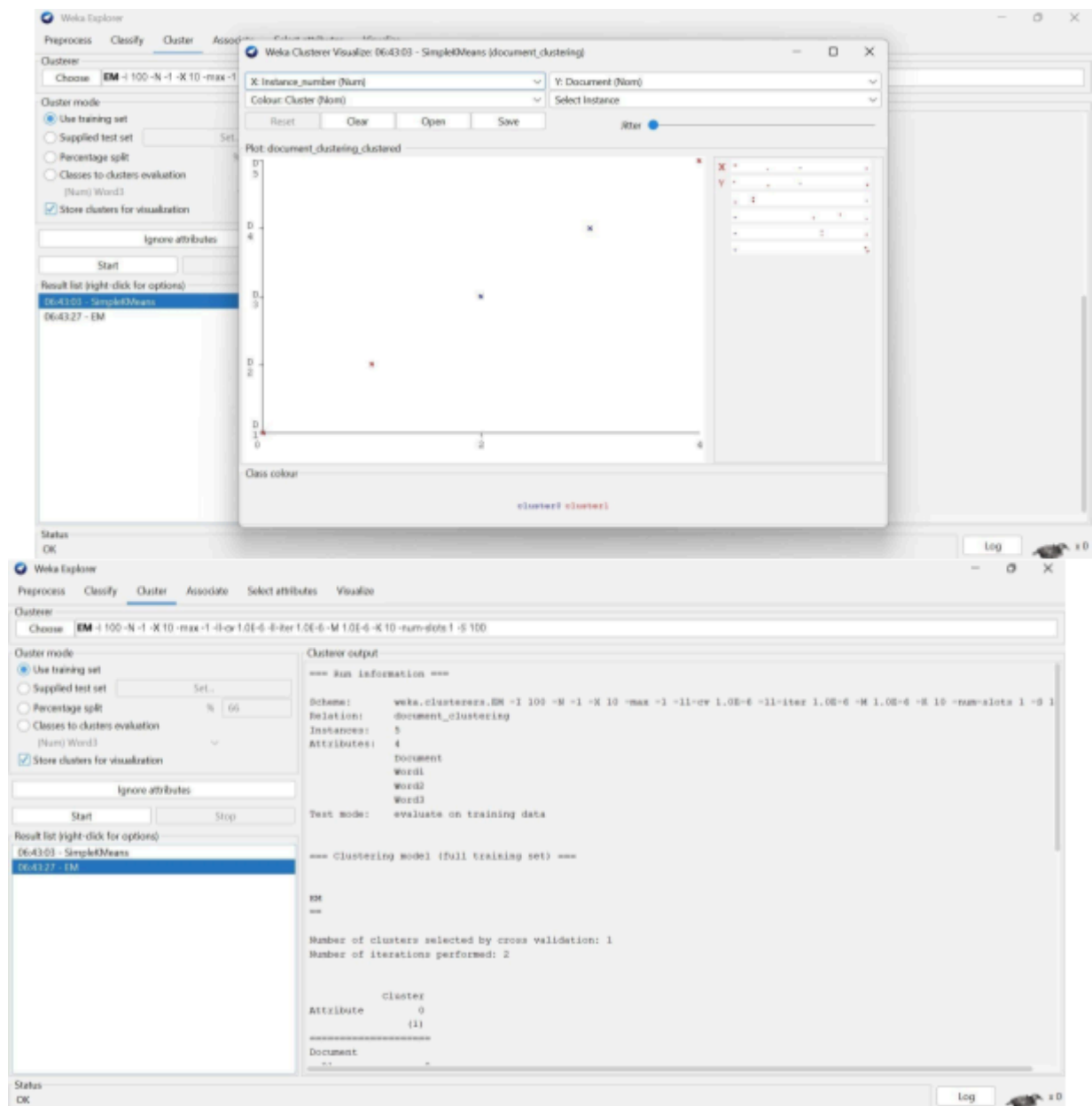
Log likelihood: -0.90915

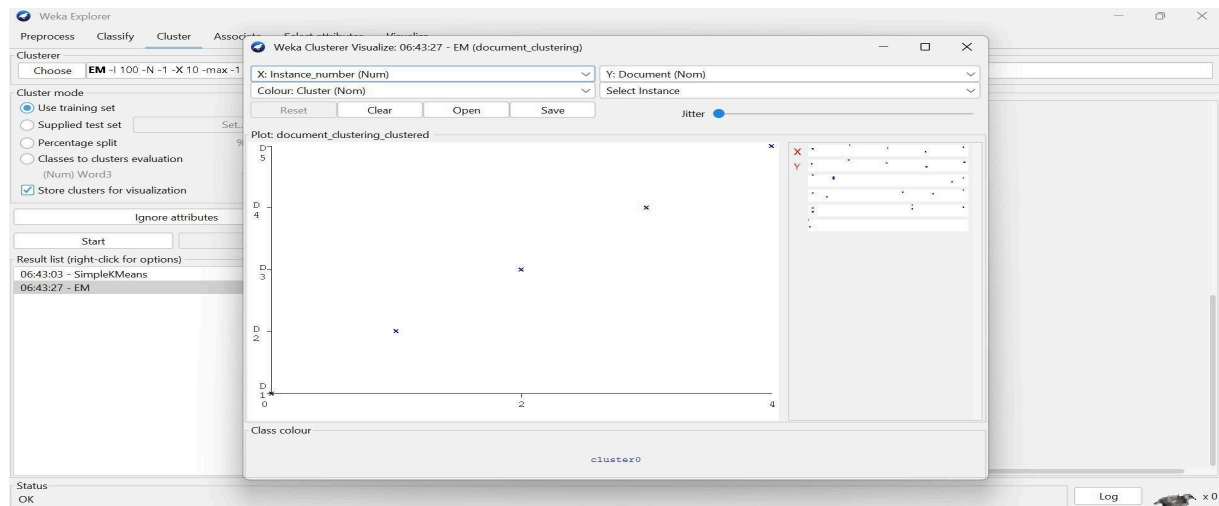
Status

OK

Log

x 0





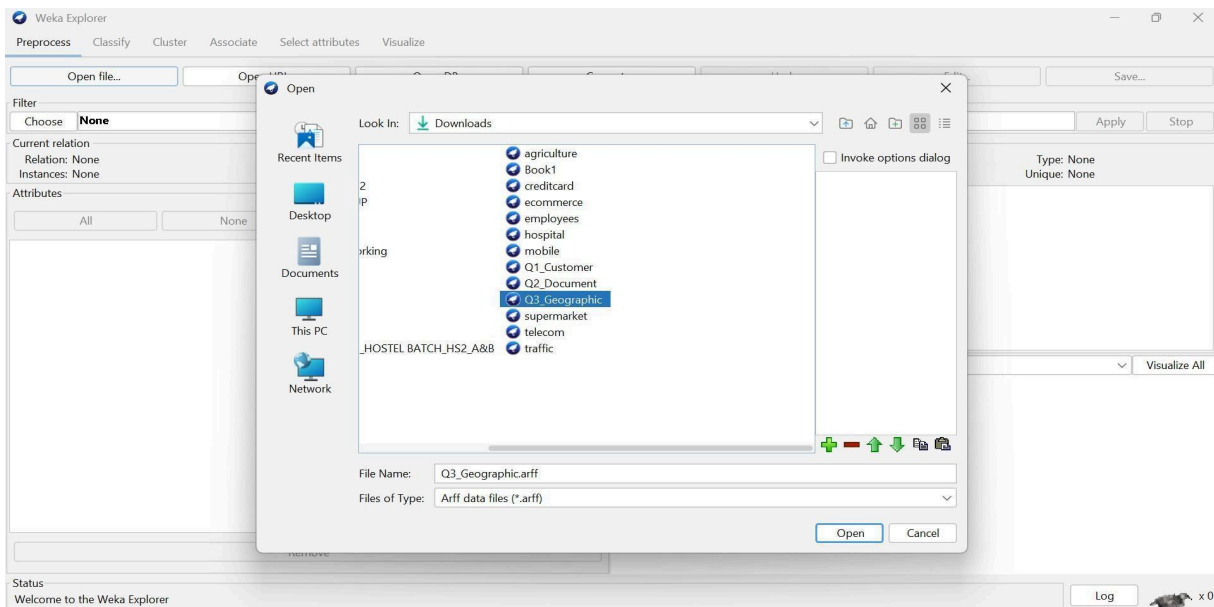
Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)



OUTPUT:

The image shows two screenshots of the Weka software interface. The top screenshot shows the 'SimpleKMeans' configuration window in 'weka.gui.GenericObjectEditor'. The bottom screenshot shows the 'Weka Explorer' window with the 'SimpleKMeans' model applied to a dataset, displaying the 'Cluster output' and 'Clustered Instances'.

Top Screenshot: SimpleKMeans Configuration

The 'weka.gui.GenericObjectEditor' window shows the configuration for the 'SimpleKMeans' algorithm. The 'Cluster mode' is set to 'Use training set'. The 'Store clusters for visualization' checkbox is checked. The 'distanceFunction' dropdown menu is open, showing options: 'ChebyshevDistance', 'EuclideanDistance' (selected), 'FilteredDistance', 'ManhattanDistance', and 'MinkowskiDistance'.

Bottom Screenshot: Weka Explorer Output

The 'Weka Explorer' window shows the 'SimpleKMeans' model applied to a dataset. The 'Cluster output' section displays the following information:

Initial starting points (random):

Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:

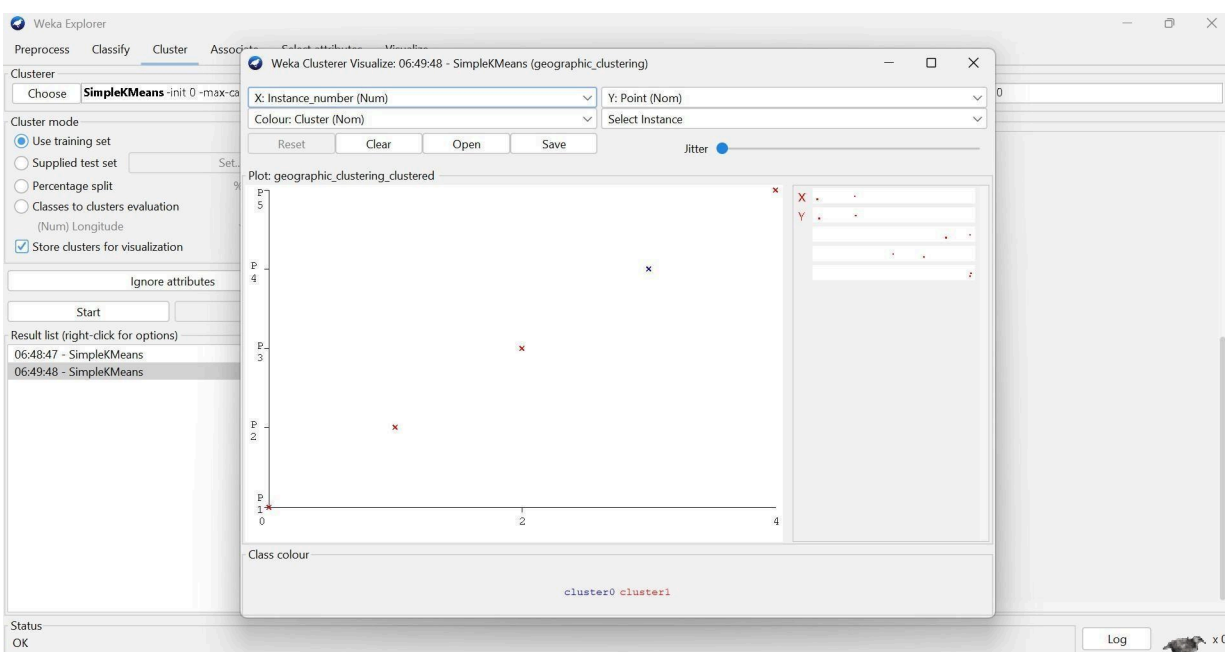
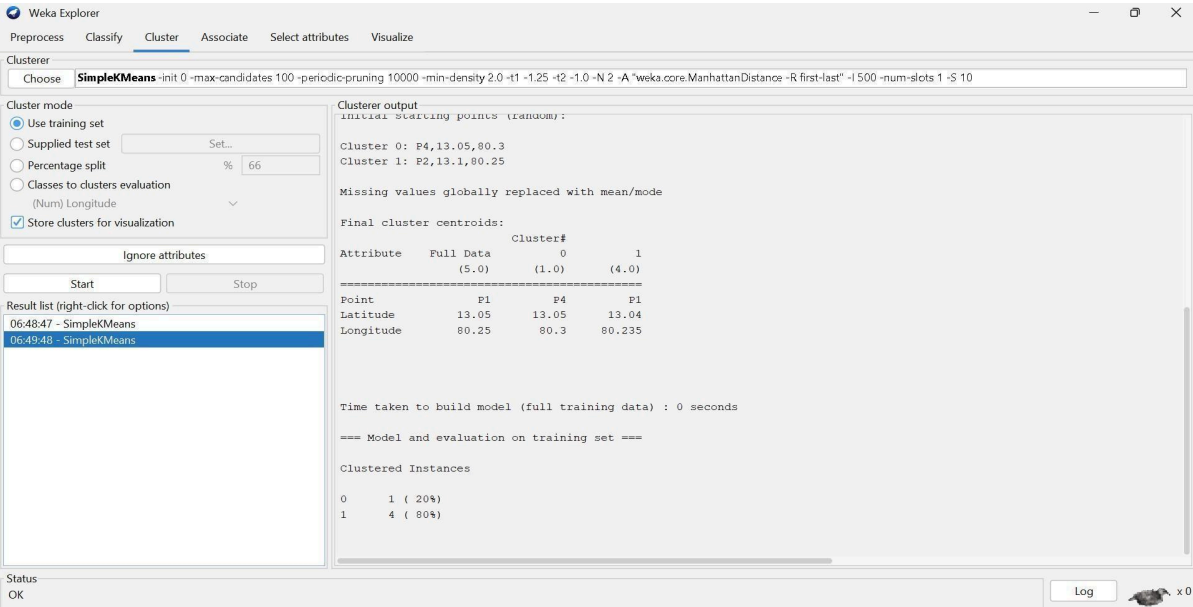
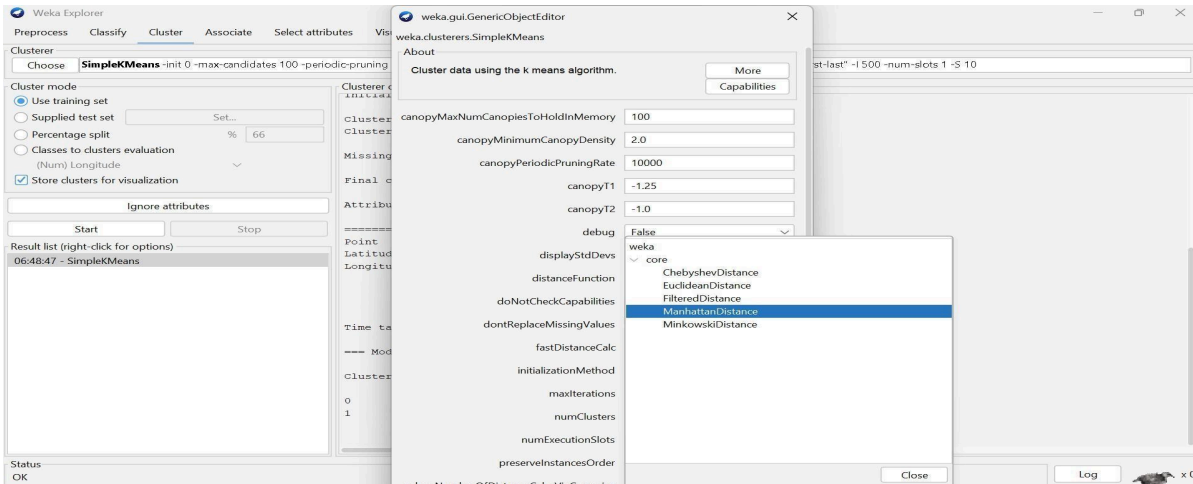
Attribute	Full Data	Cluster#	0	1
	(5.0)		(2.0)	(3.0)
Point	P1	P3	P1	
Latitude	13.04	12.985	13.0767	
Longitude	80.248	80.21	80.2733	

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	2	(40%)
1	3	(60%)



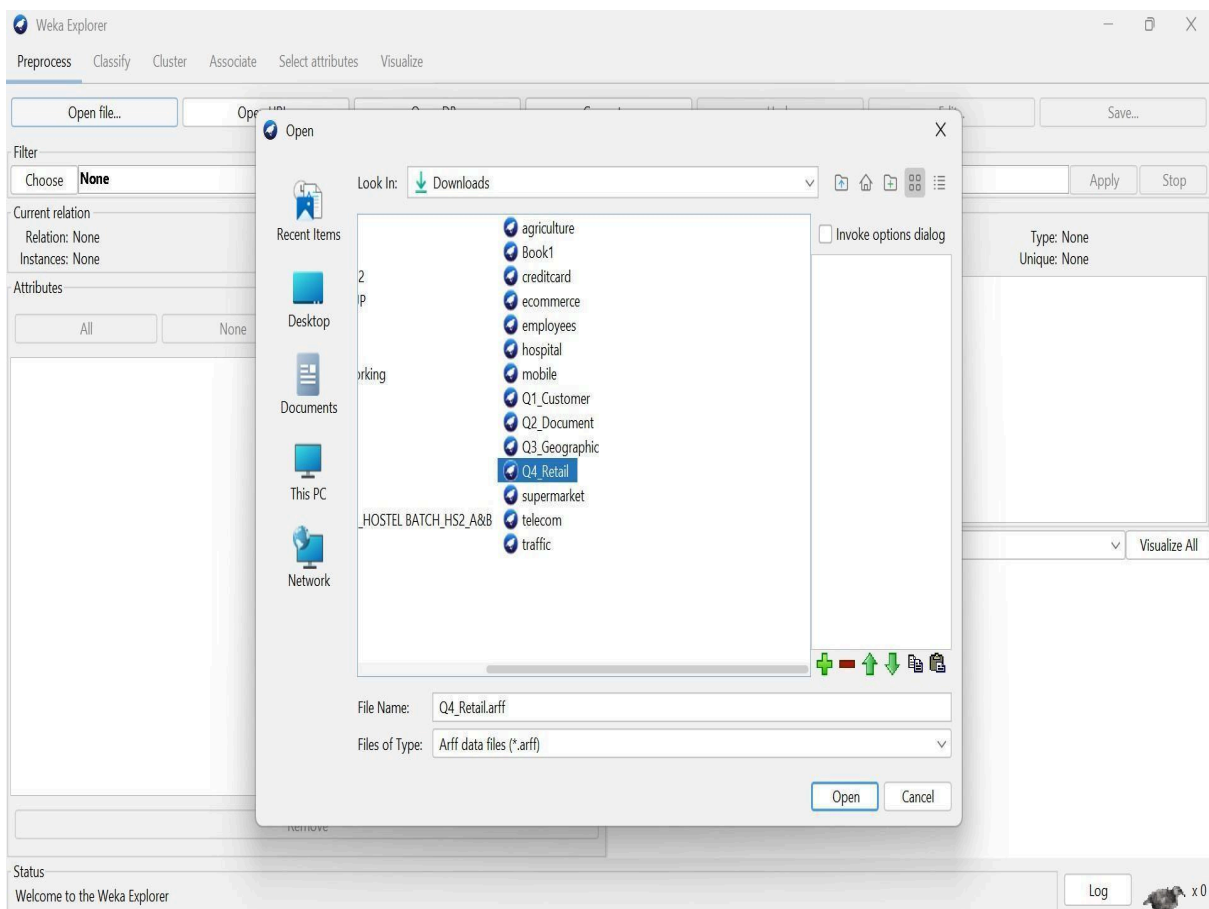
Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)



OUTPUT:

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) Frequency
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:53:39 - HierarchicalClusterer

Clusterer output

```
Relation: retail_products
Instances: 5
Attributes: 3
  Product
  Sales
  Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
(10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0    2 ( 40%)
1    3 ( 60%)
```

Status: OK Log x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans --init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) Frequency
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:53:39 - HierarchicalClusterer
- 06:54:50 - MakeDensityBasedClusterer

Clusterer output

```
Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

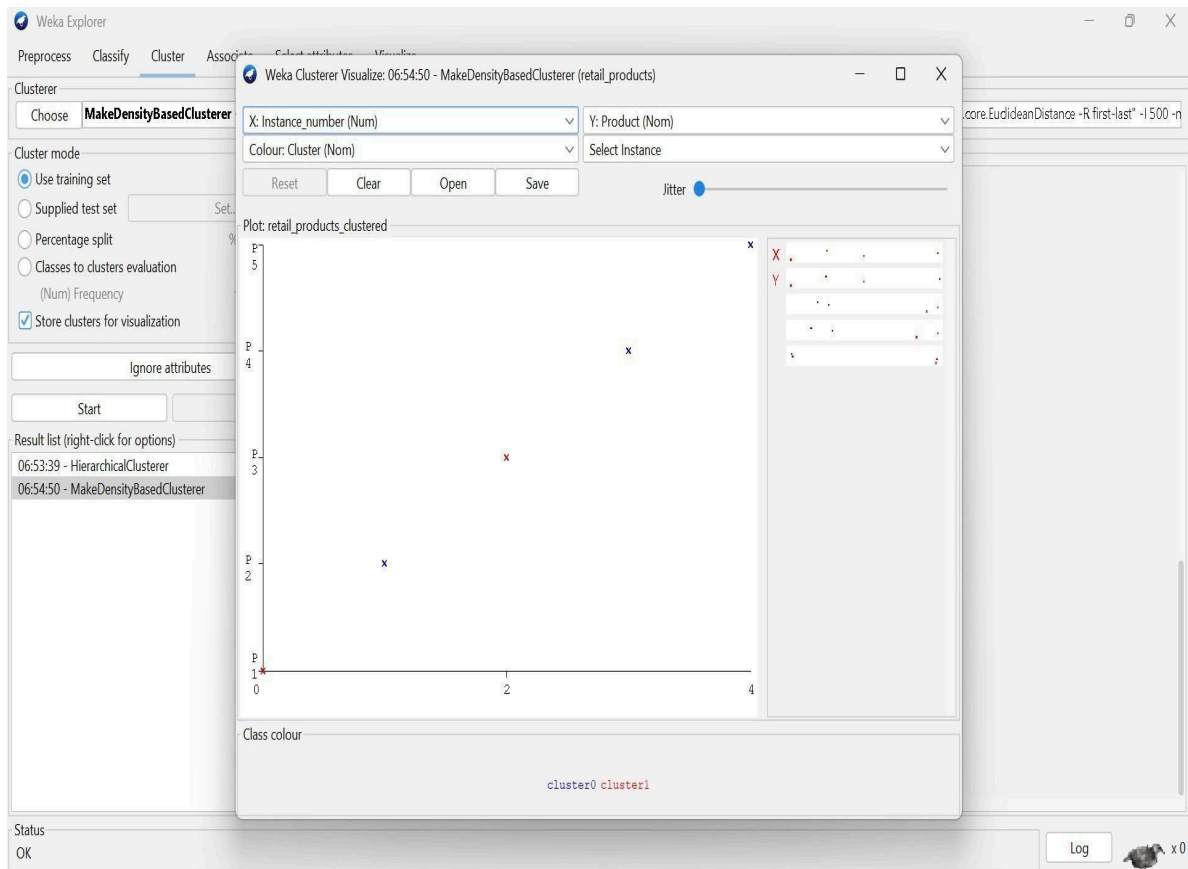
=== Model and evaluation on training set ===

Clustered Instances

0    3 ( 60%)
1    2 ( 40%)

Log likelihood: -12.6953
```

Status: OK Log x 0



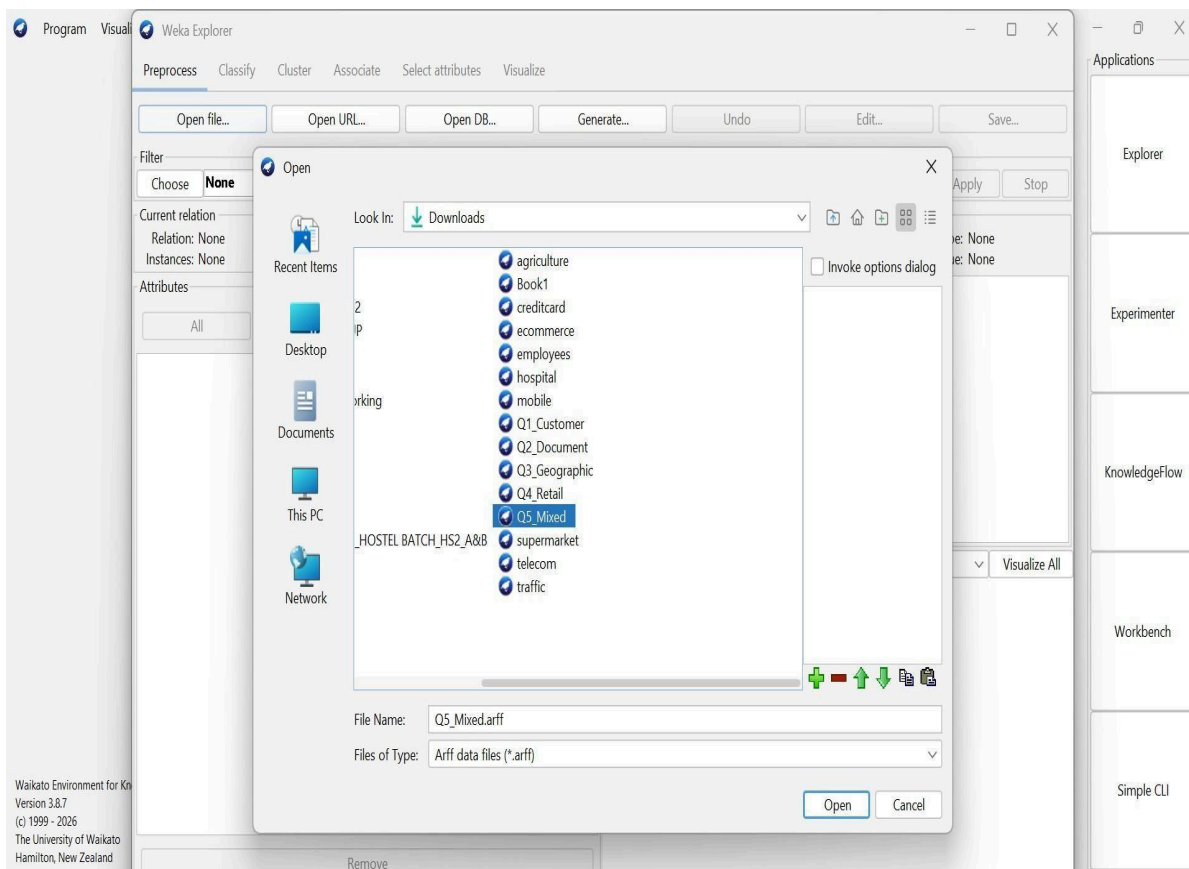
Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)



OUTPUT:

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) Rating
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:59:20 - SimpleKMeans

Clusterer output

Initial starting points (random):

Cluster 0: 250,B,3.5
Cluster 1: 300,B,3.8

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Cluster#	
	0	1
Price	424	523.3333
Category	A	A
Rating	4.22	4.6

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	2	40%
1	3	60%

Status: OK

Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer**

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split %
- ☐ Classes to clusters evaluation (Num) Rating
- ☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)

- 06:59:20 - SimpleKMeans
- 06:59:39 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:59:39 - MakeDensityBasedClusterer (mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1_clustered)

X: Instance_number (Num) Y: Price (Num)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1_clustered

Class colour

cluster0 cluster1

Status: OK

Log

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Clusterer

Choose

MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

Set...

%

66

▼

☒ Store clusters for visualization

Ignore attributes

Start

Stop

Result list (right-click for options)

06:59:20 - SimpleKMeans

06:59:39 - MakeDensityBasedClusterer

Clusterer output

Normal Distribution. Mean = 2.75 StdDev = 2.5
Attribute: Category
Discrete Estimator. Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution. Mean = 3.65 StdDev = 0.15

Cluster: 1 Prior probability: 0.5714

Attribute: Price
Normal Distribution. Mean = 523.3333 StdDev = 20.548
Attribute: Category
Discrete Estimator. Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution. Mean = 4.6 StdDev = 0.0816

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 2 (40%)
1 3 (60%)

Log likelihood: -4.6007

Status

OK

Log

x 0